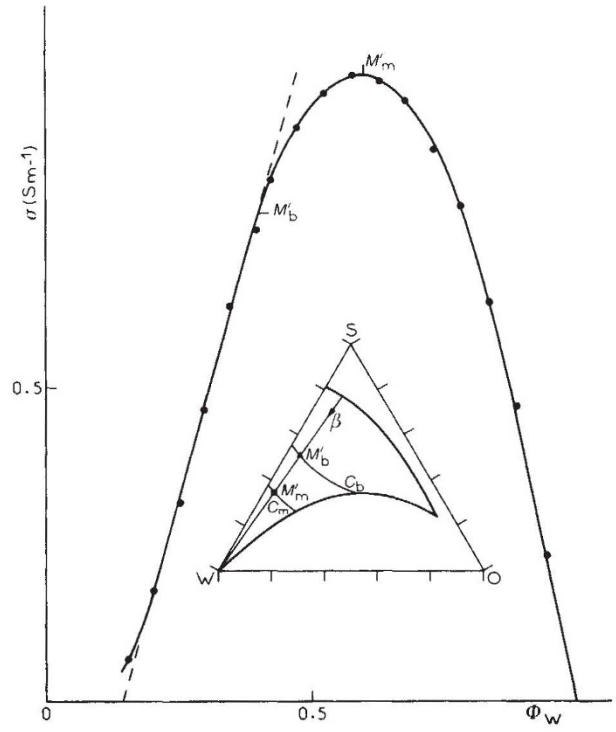


Fig. 4 Variations of σ with ϕ_w. The mass ratio r of C_6H_6 to combined SDS and 2Me-2BU is kept at 1/9, as illustrated by line β. General specifications as in Fig. 1.

These results show that the unique monophasic region of microemulsion systems of type 2 can be divided into three sub-regions labelled on Fig. 1 W/O, IZ and O/W. The boundaries between the subregions are curves C_b and C_m that are defined experimentally as conductivity transition loci. Within W/O, that is, for low water contents, the systems exhibit a percolative electrical conduction (Fig. 2) that can be accounted for by considering droplet interlinking and clustering processes in w/o-type microemulsion[15,16]. Within O/W, that is, for high water contents, the systems are made up of hydrocarbon spheres surrounded by a mixed surfactant and cosurfactant film and suspended in an aqueous phase whose conductivity decreases on addition of water (descending branch of Fig. 4). In the IZ sub-region, bounded by curves C_b and C_m, the anomalous conductivity behaviour after the percolation regime (Fig. 3) suggests that the systems are in a hybrid state which is neither w/o nor o/w. IZ can thus be considered as the phase inversion zone over which the w/o to o/w transformation occurs progressively, the systems undergoing no brutal macroscopic alterations, in contrast with the phenomena encountered in the case of type-1 microemulsion systems whose phase diagrams exhibit two disjointed monophasic areas[21,22]. These results strongly support the previous suggestion[15,16] that the model of bicontinuous structures could be applied to the phase inversion phenomenon in Winsor IV systems. It has also been proposed[29], on the basis of viscosity and conductivity data, that middle-phase microemulsions which are fluid transparent isotropic media in equilibrium simultaneously with an excess water phase and an excess hydrocarbon phase could be assimilated to bicontinuous structures. Should so-called bicontinuous structures be more than a theoretical concept, they deserve thorough investigation as they could represent an unusual type of multicomponent organized fluid in dynamical equilibrium. We are hoping to generalize the results reported here by investigating microemulsion conductive behaviour as correlated to phase diagram features and cosurfactant conformation.

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Palaeomagnetic evidence for a displaced terrain in Western Antartica

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An outstanding problem in Antarctic tectonics is the relationship between the East Antarctic craton and the late Palaeozoic–Cenozoic orogenic belts that comprise Western Antarctica[1] (Antarctic Peninsula, Ellsworth Land, Eights Coast and Marie Byrd Land; see Fig. 1*a*). Related problems are the place of Western Antarctica in reconstructions of Gondwanaland and the evolution of the Pacific margin of Gondwanaland. We present here palaeomagnetic data derived from folded Cambrian argillites that crop out in the Ellsworth Mountains which were collected during the 1979–80 austral field season. Although the early Palaeozoic palaeomagnetic field of Gondwanaland is characterized by rapid apparent polar wander, at the present level of coverage, the data are consistent with rotation of the Ellsworth Mountain block with respect to the East Antarctic craton, as postulated by Schopf[2]. This result supports a microplate nature of Western Antarctica.

The geology of the Ellsworth Mountains has been summarized by Craddock[3]. A continuous sedimentary sequence is estimated to be at least 13 km thick and to range in age from late Precambrian(?) to Permian. The outcrops display a variety of fold types with fold axis trending 310°E in the southern part to 345° in the north. The sedimentary section and style of folding have been compared with that in the Pensacola Mountains[2,4] and the Cape Fold Belt[3], and the deformation in the Ellsworth and Pensacola Mountains is related to the early Mesozoic Gondwanide orogeny[1,4].

The structural trends of the Ellsworth Mountains appear to be perpendicular to those of the Pensacola Mountains. Schopf[2]

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Table 1 Mean-site directions, statistics and pole positions from the Cambrian of the Ellsworth Mountains

Site	Sampling site Lat.	Long.	Tilt correction	N	$\bar{D}$	$\bar{I}$	k	α_{95}	Pole Lat.	Long.	dp	dm
1	79°12′S	85°54′W	Before	7	331°	33°	14	17°				
			After		13°	17°	15	16°				
2	79°09′S	85°56′W	Before	8	328°	55°	6	26°				
			After		20°	24°	6	26°				
3	79°10′S	86°10′W	Before	8	14°	18°	22	12°				
			After		20°	9°	25	11°				
4	79°12′S	86°03′W	Before	8	352°	−63°	6	24°				
			After		36°	10°	6	24°				
5	79°08′S	86°13′W	Before	10	22°	−16°	11	15°				
			After		21°	4°	11	15°				
Mean Upper Cambrian	79°S	86°W	Before	5	357°	8°	3	58°	***	****		
			After		22°	13°	52	11°	4°N	296°E	5.7°	11.2°
Pole position after rotation 82° clockwise about a Euler pole (84°S, 300°E)									1°S	18°E	5.7°	11.2°

'Tilt correction' denotes whether the parameters are before or after structural correction is applied; N, number of samples or sites from which the mean is calculated; $\bar{D}$, mean declination; $\bar{I}$, mean inclination; k, estimate of Fisher's precision parameter[17]; α_{95}, the apical half angle of the cone of 95% confidence[17]; the palaeomagnetic pole is calculated from the mean of the structurally corrected mean-site directions; dp and dm are the axes of the oval of 95% confidence[18].

suggested that the Ellsworth Mountains are allochthonous with respect to the Pensacola Mountains and placed the Ellsworth Mountains north of the Pensacola Mountains (Fig. 1*b*) so that the structural trends align; this hypothesis has since been elaborated[5–7]. Others[3,8] have suggested that the Ellsworth Mountains are autochthonous, with the structural trends continuing beneath the ice, swinging through 90° to align with the Pensacola trend.

During the 1979–80 field season, a major field programme in the Ellsworth Mountains was conducted by the US Antarctic Research Program[9], which included a systematic palaeomagnetic collection of the Cambrian to Devonian sedimentary section. Beforehand, the magnetic remanence of a suite of unoriented samples collected by expeditions during the early 1960s was investigated by stepwise alternating field and thermal demagnetization and plotting the results on the orthogonal projections of Zijderveld[10]. Promising lithologies were identified on the basis of linear trajectories on the projections[11], consistency of directions within a given sample and the physical properties of the remanence, such as coercivity and blocking temperature consistent with single-domain carriers. Haematitic argillites of the Upper Heritage Group from Pipe Peak (79°09′S, 84°14′W) were identified as promising targets because the highest blocking-temperature component of their multivector remanence had shallow inclination relative to the bedding plane. Palaeogeographical reconstructions[12] of Gondwanaland show Antarctica in an equatorial position only for the Lower Palaeozoic; the shallow inclination with respect to bedding identified for these argillites is therefore consistent with Lower Palaeozoic magnetization. They were therefore more extensively sampled over a wide area in 1979–80.

The stratigraphical horizons collected may be formally reclassified and renamed as a result of work done on this and previous expeditions (B. Spörli and G. Webers, personal communication). The sampling sites are stratigraphically within 200 m of a horizon traceable through certain structures in the northern Heritage Range. The horizon is dated to the Upper Cambrian on the basis of a trilobite fauna recovered on the flank of Pipe Peak during 1979–80 (G. Webers, personal communication). The sampling sites are spread over a 25-km^2 area. Sites 3 and 5 are in subhorizontal beds near the centre of the Pipe Peak Syncline, site 4 is on the vertically dipping, westward younging flank of this syncline, and sites 1 and 2 are in the vertically dipping eastward younging flank of the adjacent Tent Syncline.

Magnetic remanence was measured using the British prototype CCL cryogenic magnetometer and a DIGICO spinner magnetometer. Stepwise thermal demagnetizations were performed with a bench-type mu-metal shielded oven and a larger oven described in ref. 13. The results were analysed as described previously. Natural remanent magnetism (NRM) intensities range from 10^{-1} Am^{-1} to 10^{-4} Am^{-1} but are typically 10^{-3} Am^{-1}. Alternating field demagnetizations using a tumbler device resulted in only partial removal of secondary components and characteristic directions were not well resolved.

Table 1 summarizes the mean-site directions and associated statistics of the highest blocking-temperature components (650–670°C) revealed by thermal demagnetization. The mean is computed from the directions of magnetization isolated from individually oriented specimens. Figure 2 shows stereographical projections of the mean-site directions and cones of 95% confidence before and after structural correction. *In situ*

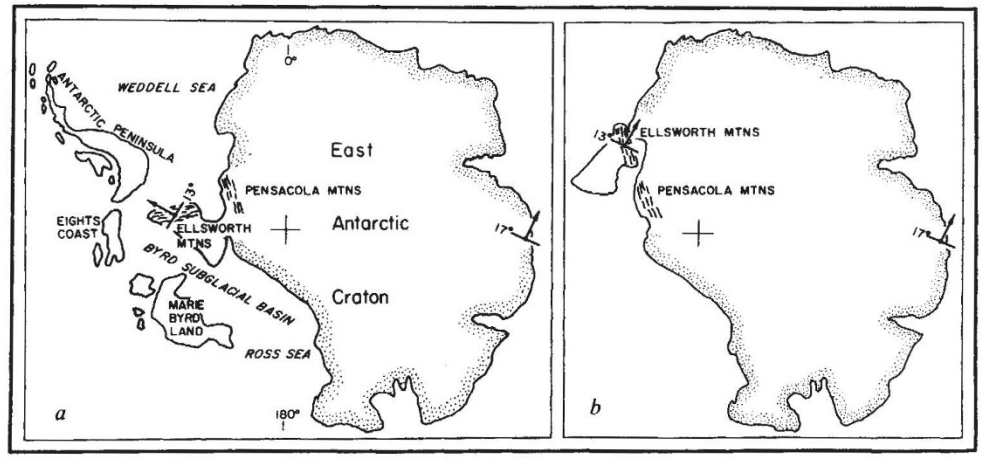


Fig. 1 *a*, Map of Antarctica showing major blocks of continental crust above sea level after ice is removed[1]. The structural trends defined by fold axes are shown for the Ellsworth and Pensacola Mountains. Mean palaeomagnetic declinations and inclinations are indicated for the Cambrian argillites of the Ellsworth Mountains and the one late Cambrian result from the East Antarctic craton[16]. *b*, Map illustrating a suggested restoration of the Ellsworth Mountain block[2]. The mean palaeomagnetic declinations and inclinations illustrate the consistency of the present data with the rotation hypothesis. The extent of the Ellsworth block must be regarded as hypothetical.

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directions are different for each structural limb, and agree, with a complete overlap of the cones of confidence, only after the correction for the tilt of the beds is applied, proving a pretectonic age of the highest blocking-temperature component. Although the positive fold test does not prove a Cambrian acquisition of the highest blocking-temperature component, we still believe that the shallow inclination of the structurally corrected remanence does imply a Lower Palaeozoic age for this magnetization.

Studies of North American rocks indicate that the Upper Cambrian was a time of predominately reversed polarity[14], although some normal polarity episodes are recorded in rocks from Australia[15]. We therefore believe that the components isolated from the Upper Cambrian argillites of the Ellsworth Mountains are probably of reversed polarity. Assuming this is the case, the palaeomagnetic south pole falls at 4°N, 296°E, approximately 70° of arc away from any known Gondwana early Palaeozoic palaeomagnetic south pole with respect to Antarctica. If the Ellsworth block is restored to a position like that advocated by Schopf[2] (Fig. 1*b*), the palaeomagnetic pole falls on Klootwijk's[15] interpretation of the Cambrian polar wander track (Fig. 3) between late middle Cambrian (poles 1, 18, 23 and so on) and Upper Cambrian (poles 28, 29) poles from Australia. The restored Ellsworth palaeomagnetic pole is also indistinguishable from the one late Cambrian pole available from the East Antarctic craton[16]. This consistency is depicted in Fig. 1, where the mean declinations of the palaeomagnetic results are compared before and after the restoration of the Ellsworth Mountains.

If the characteristic magnetizations from the Ellsworth Mountains are of normal polarity, the uncorrected south pole falls as indicated in Fig. 3, approximately 30° of arc away from the Gondwana polar wander track. This interpretation would still imply a rotation of the Ellsworth Mountain block, but in a sense opposite to that suggested by Schopf[2], which would not align the structural trends.

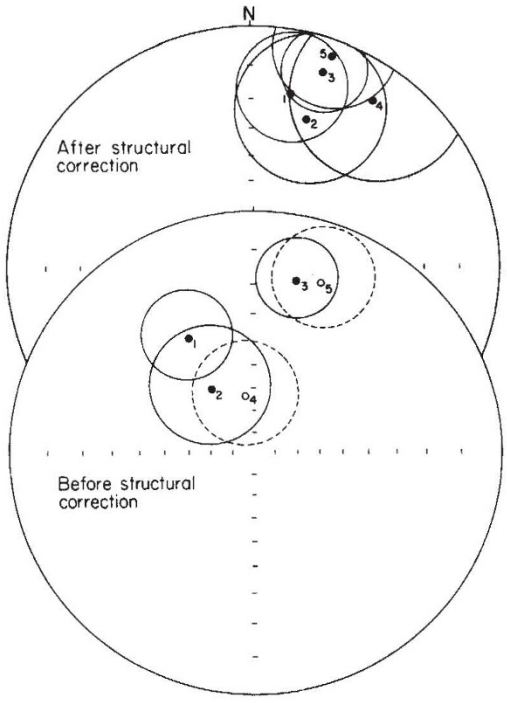


Fig. 2 Equal-angle stereographical projections of the mean-site directions with associated cones of 95% confidence, before and after structural correction. The mean directions are labelled as in Table 1. ●/○ Denote projections of the north-seeking polarity directions on the lower/upper hemisphere. The continuous/dashed circles denote the projections of the cones of confidence on the lower/upper hemisphere. The cones of confidence overlap only after the structural correction is applied.

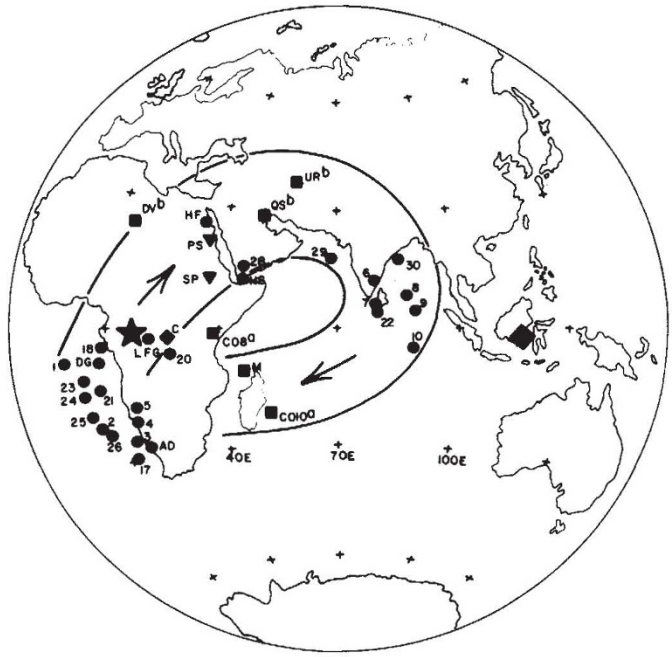


Fig. 3 Middle Cambrian to Cambro-Ordovician Gondwana polar wander path with respect to Antarctica after Klootwijk[15]. Numbered poles are the Australian poles reported in ref. 15, (a) poles are from ref. 19, (b) poles are from ref. 20, all others are from ref. 21. Pole 'C' is the one late Cambrian result available from the East Antarctic craton[16]. The star denotes the location of the mean Upper Cambrian pole from the Ellsworth Mountains after a rotation suggested by Schopf[2] is applied (Table 1). The large diamond denotes the non-rotated south pole position assuming a normal polarity. ●, Australian poles; ◆, Antarctic poles; ■, African poles; ▼, Indian poles.

Uncertainties in polarity determination and the rapid polar wander with respect to Gondwanaland limit the usefulness of the present data in deriving a precise location of the Ellsworth Mountain block in Gondwanaland. However, assuming the characteristic magnetizations are of reversed polarity, the restoration suggested by Schopf[2] is consistent with these data. Our results support interpretations that Western Antarctica is composed of displaced terrains which have moved with respect to the major plates.

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Evidence for a late Wisconsin glaciation of the Weddell Sea

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The huge ice shelves in West Antarctica—the Ross and Filchner/Ronne Ice Shelves—have probably extended out on the continental shelf during the late Wisconsin[1]. Previous discussions, which have focused on the Ross Sea, have suggested (1) that the ice extended across the whole continental shelf[2,3] or (2) that there was only a minor expansion[4]. Here we present sedimentological data from the Weddell Sea (Fig. 1) which suggest that a late Wisconsin grounded ice sheet extended to the shelf edge. The evidence includes a recent thicker ice in Ellsworth Mountains at the head of the Filchner/Ronne Ice Shelf[5]. This thickening would lead to an expansion of the inland ice sheet over the continental shelf, filling up the Weddell Sea embayment.

Geological fieldwork during the 1976–77 and 1978–79 Norwegian Antarctic Research Expeditions provided the following observations.

(1) Sediment samples (60 stations) show that the continental shelf is covered by stiff pebbly mud (till deposits), in places overlain by a thin veneer (<2 m) of soft pebbly mud (glaciomarine deposits). From the upper slope only glaciomarine sediments are recovered (Fig. 2). A similar lithology is described by Anderson *et al.*[6] (23 cores).

(2) Shallow seismic reflection (sparker) records show a veneer of unconsolidated sediments which rapidly thickens seawards off the shelf edge[7] (G. Maisy, personal communication).

(3) ^{14}C dating of shell fragments of the till, together with the age and distribution of the glaciomarine sediments, suggest a late Wisconsin age for the advance of a grounded glacier to a water depth[8] of 500 m.

The stiff pebbly mud is characterized by poor sorting, homogeneous texture and variable content of reworked fossils and significant amounts of Mg- calcite and cristobalite[9]. The degree of compaction—measured by cone—gave values in the range of 50–120 kN m^{-2} on core samples from the shelf edge. Similar values have been recorded on late Wisconsin till on the Norwegian shelf[10]. The compaction of the stiff pebbly mud may have been caused by sediment loading. However, it is unlikely that submarine erosion has been able to remove completely a substantial overburden of glaciomarine–glacial deposits. In the latter case, a lag of coarser-sized material would be present, as observed in other glaciated shelf areas which have later been reworked subaqueously[11]. The values obtained for compaction show that the deposit is overconsolidated, which, in addition to

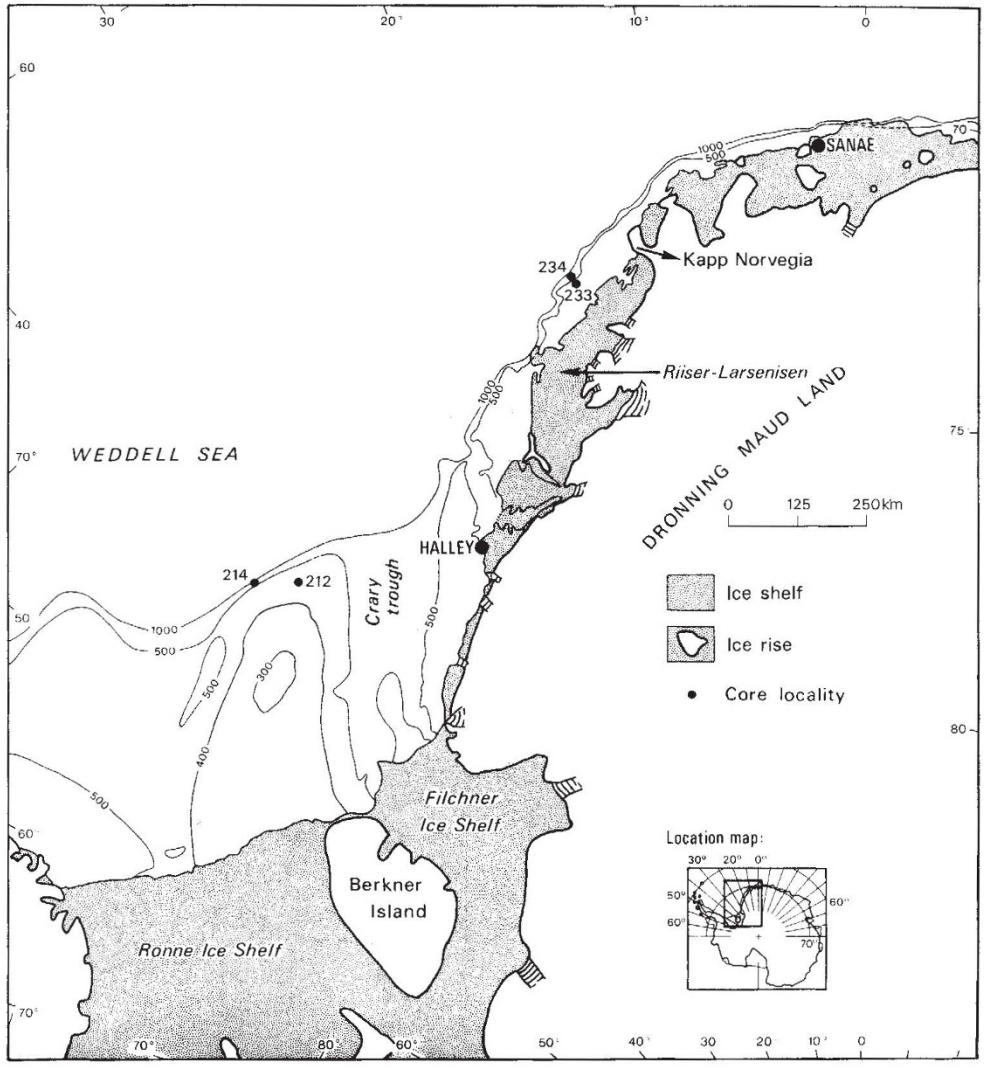


Fig. 1 Map showing the eastern part of the Weddell Sea and localities of cores shown in Fig. 2.

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